

Ikigai -- A Biologically Grounded Digital Organism: Development Report, Day 77

Mura ALife Labs (Formerly Hitoshi AI Labs) -- NeuroSeed Project

Author: Prince Siddhpara, Founder -- Mura ALife Labs (Formerly Hitoshi AI Labs) **Date:** June 20, 2026 **Project:** NeuroSeed **Subject:** Day 77 -- 23 Packs Shipped In One Day, Pack 274 v2 Yes/No Coverage Gap Closed, Pack 277 Paper Draft v2 Reframed With RHC Reattribution (Kymn-Kleyko-Frady-Bybee-Kanerva-Sommer-Olshausen 2023, Full 7- Author Citation), Pack 278 v0+v1 Legacy Refactor (OPERATOR_LEXICON Purged, Phase 5 Deleted, Day 73 Commitment Closed), Pack 279 Compact Cache 6.78x Shrink (288 B -> 42 B Per Entry), Pack 280 Vectorize recall_action 28s -> 77ms = 364x (THE BOTTLENECK UNBLOCK), Pack 281 Parallel Teacher Batch 1.25 q/s -> 10.68 q/s = 8.56x, Pack 282 LMDB On-Disk Cache 44 B SSD/Entry With 2.4 us Warm Read + Pack 282.5 Production Wiring Via Env Var, Pack 283 Atomic-vs-Compositional Routing Skips Math Cache Writes, Pack 284 Operator Coverage Complete 11/11 (Lexicon Residue Purged), Pack 285 v0 SBC Encoder 10K Discrimination At d=400 (Frady-Sommer Wall Broken) + Pack 285.5 Action Codebook Research Primitive, Pack 286 Paper v3 LaTeX + 5 Figures + Kanerva Email Drafted, Pack 287 v0+v1 KL-RPE Continuous Graded Fire Signal (vLLM Logprobs End-To-End Validated), Pack 288 Parallel Particle-Swarm Cleanup 6.6-10000x, Pack 289 Trait-Conditional Dopamine + Last-Word Parser Fix (Greece + Bolivia Bugs Closed), Pack 290 Topic-Conditional Sleep Replay (Pack 271 Revived Without Pollution), Pack 291 Multiplicative * Binding Per Kymn §4.1.2 + Pack 291.5/291.6/291.7 Wiring And Division Operator (FULL SUBSTRATE-NATIVE ARITHMETIC RING CLOSED: add + sub + mul + div), End-To-End Teach Pipeline 170s/Fact -> 0.55s/Fact = ~310x, Day 77 Bench 30/30 (21 Capitals + 4 Add/Sub + 5 Mul/Div), Cumulative Day 76 + Day 77 Substrate-Only ICL Recall Holds At 100% **Classification:** Research Document -- Computational Neuroscience / Artificial Life / Cognitive Systems Engineering

1. Objective

Day 76 closed at six escalating wins -- the cleanup wall bypassed via the Kanerva-2022 anchor-action cache, OPERATOR_LEXICON deleted from cat-3 dispatch (Pack 263), 5 new capitals teacher-taught and persisted (Pack 276), substrate-only 25/25 bench at 100 percent, 5-file deep-research handoff returned with two converging surveys, codebase cleanup of 4 dead modules and 11 failed experiments. Pack 274 v1 had subsampled 240 NO chains to 200 and dropped the Germany-tokyo pair into the discarded 40; the wiring smoke caught it as 1 miss out of 11 right before the planned compact.

Day 77 has eight mandates.

First: close the Pack 274 yes/no coverage gap. Re-bootstrap NO chains without the subsample so every country-pair has all three phrasings in the cache.

Second: harden the production cleanup path. Pack 280 vectorize is the single biggest latency unblock pending in the locked queue -- 28 seconds per `recall_action` call against a 63K-anchor substrate. The Pack 276 dopamine retry loop fires this six times per fact teach; everything downstream of teaching scales with this number.

Third: finish the Day 73 hardcoding-removal commitment. Pack 263 took out `math_eval's _ADD_TOKENS`. Day 73 named four more removal targets (`reasoning_engine.py OPERATOR_LEXICON`, `verifier.py operator lexicon`, `conversation.py _safe_eval + IntentRouter math branch`, `t2s_compiler.py`). Pack 278 needs to ship them.

Fourth: close the substrate-native arithmetic ring. Pack 254 RHC ships add and sub via Hadamard product on bundled FPE phasors. Kymn et al. 2023 section 4.1.2 defines a separate multiplicative binding operator $\star$ via per-modulus modular inverses. Implement it on `ResidueHDC`, wire it into `MathEval.substrate_arith`, then add division via the same arithmetic-verifier path.

Fifth: prepare the cache for brain-scale. The Day 76 close articulated the compact (50 bytes/entry) and LMDB (40 bytes SSD/entry) scaling levels as projections; Day 77 needs to ship both as working code with bench gates.

Sixth: continuous reward signal. The bool substring match used in Pack 276 dopamine is the biological violation Schultz 1997 specifically measured against (real VTA firing is graded). Build KL-graded fire signal that consumes vLLM teacher logprobs.

Seventh: revive the failed Pack 271 sleep cycle along CHARLES-D classifier-free guidance lines so the organism can consolidate between teaching sessions without polluting state archetypes across topics.

Eighth: rewrite the paper draft v2 with proper RHC attribution to Kymn-Kleyko-Frady-Bybee-Kanerva-Sommer-Olshausen 2023 (the v1 framing claimed RHC as a NeuroSeed contribution, which it is not), then ship a LaTeX version with figures and a Kanerva outreach email drafted for end-of-day handoff.

Day 77 ended at twenty-three packs shipped, no failed-and-reverted experiments, one external dependency installed (Tectonic 0.15.0 single-binary LaTeX engine), one `serve.py` redeploy on the 3090 to expose vLLM logprobs, the substrate-native arithmetic ring closed (add + sub + mul + div, zero hardcoded lexicon), the cleanup latency floor dropped 364x, the per-fact teach pipeline ~310x faster end-to-end, the cache representation reduced 6.78x in memory and prepped for on-disk LMDB scaling to 10^9 facts at 42 GB SSD, the substrate-only ICL recall holding at **30/30 = 100% on 21 world capitals + 4 add/sub + 5 mul/div word problems**, and paper draft v3 in LaTeX with 5 figures plus a Kanerva email queued for re-read.

2. Pack 274 v2 -- Yes/No Coverage Gap Closed

Pre-compact smoke at the Day 76 close caught one miss out of eleven on the wiring bench: "Is the capital of Germany Tokyo" returned 'fliers'. The cache hoist fix that landed Germany->Berlin in the cache (general_reasoner.py lifted the deterministic anchor lookup ahead of the substrate state_sim gate) could not help on Germany-tokyo because the corresponding NO chain was not in the cache.

The root cause: Pack 274 v1 bootstrap generated 240 NO chains ($16 * 15$ country permutations * 3 phrasings = 720 total, then subsampled to 200 via `random.shuffle(seed=274)`). The Germany-tokyo pair fell into the discarded 40.

Pack 274 v2 dropped the subsample. All 720 NO chains absorbed alongside the 48 YES chains for 768 total. Cache grew from 317 to 837 entries. Germany-tokyo probe via `_stable_anchor` confirmed `[('no',)]` post-bootstrap. organism.ikg disk size unchanged at 182.5 MB (substrate dominates; cache 235 KB -> 35 KB pre-Pack 279 already invisible). Backup at `organism.ikg.bak_pre_pack274v2_20260619_211010`.

Lesson: `random.shuffle` + slice subsample on a combinatorial bootstrap is a bug magnet. Write all permutations; let the cache writer dedupe.

3. Pack 277 -- Paper Draft v2 Reframing And RHC Reattribution

The v1 draft `papers/draft_rhc_d400_saturation_2026_06_18.md` framed Residue Hyperdimensional Computing as an empirical contribution of this paper. That attribution is wrong: RHC is the central construction of Kymn-Kleyko-Frady-Bybee-Kanerva-Sommer-Olshausen 2023 (arXiv:2311.04872v1, seven authors including Pentti Kanerva himself and Bruno Olshausen, Redwood Center for Theoretical Neuroscience). Prince surfaced the misattribution mid-session: "wait, RHC is already existing paper, it's now our inventions."

Pack 277 reworked the paper:

1. Retitled to "Bypassing the d=400 Cleanup Wall: Associative-Cache Substrate, Organism-Style Reinforcement Learning, and Emergent Grounding for a 192 MB Hyperdimensional Organism." Dual filename: original `draft_rhc_d400_saturation_2026_06_18.md` preserved for git history; new canonical `draft_cache_bypass_organism_rl_2026_06_19.md` sits alongside.
2. Abstract restructured to (a) credit Kymn et al. 2023 + Frady-Kymn-Anandkumar 2023 for RHC, (b) list four novel contributions clearly: emergent grounding, cache bypass, organism-style RL, end-to-end empirical lift.
3. §2 Background expanded to cite Kanerva 2022 "Variations of superposition vector memory and the SDM" (Frontiers in Neuroscience 16:867568) as the prior basis for the cache primitive, plus Schultz 1997 and Aston-Jones 2005 as biological grounding for the RL loop.
4. §4 retitled "Residue Hyperdimensional Computing at 10^9 capacity (prior work, applied)" with an explicit Attribution paragraph: "We do not claim the encoding scheme itself." The seven-author citation appears in five places (abstract, intro, background, §4 attribution, references).

5. New §7 "Bypass via Anchor-Action Cache + Organism-Style Reinforcement Learning" inserted between §6 Discussion and the existing Future Work (renumbered §7 -> §8, Conclusion §8 -> §9, References -> §10). Five subsections covering the Kanerva-2022 lookup primitive, storage vs discrimination capacity decoupling, Cat4Dopamine RL loop, DM-EA parallel discussion (Faldor + Cully 2024, Zhang HADES 2024), and limitations.
6. References expanded with Kanerva 2022, Schultz 1997, Aston-Jones 2005, Williams 1992 REINFORCE, McClelland-McNaughton-O'Reilly 1995 complementary learning systems, plus the two DM-EA cites.

The reattribution gains weight for the Pack 286 Kanerva outreach: NeuroSeed extends his own RHC paper at d=400 / 192 MB rather than claiming the construction.

4. Pack 278 v0 + v1 -- Legacy Refactor (Day 73 Commitment Closed)

Pack 278 v0 (morning) added deprecation headers to four legacy modules without breaking imports:

`reasoning_engine.py` (OPERATOR_LEXICON), `verifier.py` (GSM8K free functions), `t2s_compiler.py` (Day 72 vision pivot), `conversation.py` (`_safe_eval` + IntentRouter math branch). Phase 5 in `integrate.read_statement` (`legacy self.re.read_statement(text)`) marked dead in a comment block.

Bench gate after each header: load OK, France->paris, 5+3=8. 15+ call-sites still touched `self.re/self.reasoner/self.verifier/self.t2s` so full file deletion was deferred to v1.

Pack 278 v1 (afternoon) shipped the actual surgery:

1. `reasoning_engine.py` OPERATOR_LEXICON replaced with empty dict (140 hardcoded verb-stem tokens deleted: SET / ADD / SUB / MUL / DIV). All `tok in OPERATOR_LEXICON` checks degrade to False silently.
2. `integrate.py` Phase 5 block removed entirely from `read_statement`. Substrate now handles math via Pack 254 RHC + Pack 291 multiplicative *, capitals via Pack 273 cache + Pack 276 dopamine, yes/no via Pack 274 v2 cache.
3. The four legacy modules retain their v0 deprecation headers but stay in place (full file deletion = Pack 278 v2 if/when `integrate.py` rewires the lazy import sites). The 15+ remaining call-sites are not on the 25/25 production path.

Bench gate after v1: cache class CompactAnchorCache (Pack 279 effect, see §6), 5/5 cache probes OK, math 4/4, capitals 21/21, total 25/25, save+reload survives. No regression. The Day 73 commitment ("delete OPERATOR_LEXICON hardcoding from the production path") is closed.

5. Pack 279 -- Compact Cache (6.78x Shrink)

Pack 273 + Pack 274 stored the anchor-action cache as `dict[str_anchor, list[tuple[str]]]`. String anchors like `'__state_701357096285'` consume 70 bytes; list-of-tuple values like `[('paris',)]` consume ~170 bytes; dict slot overhead ~32 bytes. Profiled per-entry footprint: 288 bytes (close to the 270 projected at Day 76 close). At 10^6 facts that footprint is 270 MB -- exceeds the 192 MB substrate cap.

Pack 279 swaps the representation to `dict[int64, bytes]`:

- Key: 8-byte int (the raw blake2b digest used to produce the original string anchor).
- Value: utf-8 encoded action token bytes (~10 bytes average for single-word answers).
- Last-wins semantics on duplicate anchor writes (matches Pack 273 runtime read pattern which already took the LAST entry of the value list).

`CompactAnchorCache` ships in `ikigai/cognition/cat4_compact_cache.py` with a back-compat shim: `__getitem__ / get` accept string anchors and return list-of-tuples, so `general_reasoner` and `cat4_dopamine` require no changes.

Production smoke: load existing organism (legacy dict cache), automigrate on attach via sample-type sniff (bytes -> Pack 279 state, list-of-tuples -> legacy migrate), 25/25 bench preserved, save+reload survives with compact persistence. Measured per-entry: 42.5 bytes = 6.78x shrink. Projected: 1M facts = 40.5 MB (under 192 MB cap), 1B = 40.5 GB (LMDB territory).

`integrate.py` `cat4` lazy property does the sniff-and-migrate; `_sync_cat4_cache_for_persist` writes the compact `dict[int, bytes]` to `_cat4_anchor_actions_cache` for pickling. `organism.ikg` disk size unchanged at 182.5 MB (substrate dominates; cache 235 KB -> 35 KB is invisible). The win is at scale.

Backup: `organism.ikg.bak_pre_pack279`.

6. Pack 280 -- Vectorize recall_action (28 s -> 77 ms = 364x)

The Day 76 close articulated this pack as "the single biggest latency unblock pending in the locked queue." Pack 280 stacks all stored state hyper-vectors into a `(N_anchors, d)` `complex64` matrix cached on the `Cat4Absorb` instance and replaces the per-anchor Python loop in `recall_action` with one `matmul` plus an `argpartition` top-K selection.

Implementation:

1. New instance attrs `_pack280_recall_states`, `_pack280_recall_bounds`, `_pack280_recall_anchors`. Built lazily on first `recall_action` after invalidation.
2. `_pack280_build_recall_cache` iterates `pair_role` anchors, calls `mr.recall(a, state_role) + mr.recall(a, pair_role)` for each, stacks the results.
3. New `recall_action` body: `single states.conj() @ query_state matmul -> per-anchor sims, argpartition -> top-K, per-row action_est unbinding for the returned tuples`.

4. Incremental update on `absorb_chain`: for each new `(state_hv, bound_norm)` write, append one row to each matrix in place (existing anchor overwrite via index lookup; new anchor vstack). Never invalidates the full cache during teaching.

Smoke results:

- Cold rebuild (one-time per session): 142.6 s for the production 63,917-anchor substrate.
- Hot avg over 4 queries: 76.8 ms, range 65-87 ms.
- Speedup vs old 28 s loop: 364x.
- Memory cost: 390 MB (195 MB states + 195 MB bounds at N=63,917). Under the 1 GB cache ceiling.
- Incremental absorb path: new anchor recall returns `top_sim=1.000` immediately, no re-cold.

The cold cost is not persisted to `organism.ikg` -- session start pays it once. Sidecar disk-cache for `_pack280_recall_states` is a potential Pack 280.5 if startup latency becomes user-visible. Per-fact teach time drops from ~3 min (Pack 276 dopamine retry running 6 cleanup iterations at 28 s each) to ~0.5 s; the 1M-fact teach pipeline now fits in a single 3090's overnight budget.

7. Pack 281 -- Parallel Teacher Batch (8.56x Throughput)

`TeacherOracle.ask` previously called `_batch([prompt])` -- one HTTP request per query. Sustained at ~1.25 q/s against the 3090 vLLM server. Pack 281 adds `ask_batch(queries, batch_size=16)` returning `dict(query, answer):`

1. Pre-filter cached queries -> serve locally, no HTTP.
2. Build prompts list for uncached.
3. Single `_batch(prompts)` call, vLLM's continuous batching parallelizes inference across all prompts in the request.
4. Parse `items[i]['text']` per query, update cache, return.
5. Chunks `N > batch_size` automatically.

`teach_with_oracle` updated to pre-fetch all truths via `ask_batch` before entering the dopamine retry loop, so the oracle leg never re-hits HTTP during retries.

Real-teacher smoke on 10 fresh capital queries:

- Sequential 10 single asks: 8.0 s wall, 1.25 q/s.
- Batched 10 in one HTTP call: 0.9 s wall, 10.68 q/s.
- Speedup: 8.56x.
- All-cached shortcut (10 queries already in cache): 0 ms.
- Mixed cached + fresh (1 cached + 1 fresh): only the fresh one hits HTTP.

Combined with Pack 280, end-to-end Pack 276 fact-teach pipeline:

- Before: oracle 0.8 s + 6 cleanup iters x 28 s = 170 s/fact.
- After: oracle 0.09 s + 6 iters x 77 ms = 0.55 s/fact.
- End-to-end: ~310x faster.

Note one observed parser bug: "Bolivia" -> 'la' (first-word grab cut 'la paz'). Same class as the Day 76 Greece bug. Pack 289 fix queued.

8. Pack 282 + Pack 282.5 -- LMDB On-Disk Cache

Pack 282 ships `LMDBAnchorCache` in `ikigai/cognition/cat4_lmdb_cache.py` with the same dict-like API as `CompactAnchorCache`. `int64` big-endian key (8 bytes) -> bytes value (utf-8 token, ~10 bytes). Bulk `update_from_compact` and `update_from_dict` migrators for batched single-txn import.

Standalone smoke:

- 837 entries migrated from `CompactAnchorCache` in <0.01 s.
- Read warm: 2.4 us (target was 50 us per Day 76 projection -- 20x faster).
- `data.mdb` actual data 36 KB (`psize * used_pages`, not the 1 GB `map_size` pre-allocation stub that the initial `disk_bytes()` incorrectly reported).
- Per-entry SSD: 44 bytes.
- Projections: 1M facts = 42 MB SSD, 1B = 42 GB SSD (matches Day 76 close cache-scaling math).
- Persistence: `close + reopen readonly` survives.

Pack 282.5 wired LMDB into production behind an opt-in env var. `Cat4Absorb.__init__` reads `NEUROSEED_LMDB_CACHE`; when set to a directory path, swaps `CompactAnchorCache` for `LMDBAnchorCache` (soft-falls back to compact on `lmdb` import failure, logged to `stats.lmdb_fallback`). `integrate.py` `cat4` lazy property now handles three load paths:

1. LMDB live + persisted state -> bulk-load via `update_from_compact / update_from_dict`.
2. Compact bytes -> `from_persist_state`.
3. Legacy dict -> `migrate_dict_cache`.

`_sync_cat4_cache_for_persist` skips the LMDB snapshot (LMDB self-persists at its sidecar dir; `organism.ikg` keeps a tiny empty attr).

Production smoke with `NEUROSEED_LMDB_CACHE=c:/temp/neuroseed_pack282p5_lmdb` set: load organism, 837 entries automigrated into LMDB (36 KB), 25/25 bench preserved, `close + reopen readonly` survives. `organism.ikg` unchanged at 182.5 MB.

`pip install lmdb (v2.2.1)` succeeded inside the project `.venv`.

9. Pack 283 -- Atomic vs Compositional Routing

The cache is for the vocabulary of facts; the substrate is for the grammar of composition. Math queries cost nothing to recompute via Pack 254 / Pack 291 RHC at ~50 ms per query -- caching them duplicates substrate behavior without semantic gain, and at 10^9 -fact scale every avoided cache entry buys back ~44 bytes SSD (Pack 282).

Pack 283 adds `is_compositional_query(text)` in `cat4_dopamine.py` -- True when tokenization contains any of `{+ - * / =, plus, minus, times, multiplied, product, into, divided, over, sum, difference, quotient}` OR contains two or more pure-number tokens. `_dopamine_reinforce` gates on this: compositional queries skip `absorb_chain` + cache write, increment `cache_writes_skipped`.

Classifier unit tests 11/11. Production smoke with 6 mixed teaching pairs (3 atomic + 3 math): cache grew +3 not +6, `cache_writes_skipped=3`, all 4/4 math still substrate-native via Pack 254 / 291, Day 76 regression 5/5 including newly-taught Iceland + Mongolia atomic facts. Vocabulary vs grammar separation enforced.

10. Pack 291 + 291.5 + 291.6 + 291.7 -- Full Substrate-Native Arithmetic Ring

Pack 291 implements multiplicative $*$ binding per Kymn et al. 2023 §4.1.2. New methods on `ResidueHDC`:

- `_ensure_antibases()` -- precompute per-modulus modular-inverse anti-base phasor arrays via `pow(uu, -1, m)` per element. Rejects composite moduli with `ValueError` (the modular inverse exists only when $\gcd(u, m) = 1$; for prime m every nonzero u qualifies).
- `encode_factored(x)` -- per-modulus list of HVs, length K (skips the bundling step in `encode`).
- `bundle_factored(factors)` -- recover bundled HV by per-modulus Hadamard product.
- `mul_factored(fa, fb)` -- the $*$ operator. Per modulus: extract residues r, s from each phasor via `angle * m / (2\pi)`, compute $(r*s) \bmod m$, multiply by the anti-base residue `v_j` so the per-element u_j^2 factor cancels back to $u_j * x_1 * x_2$.
- `multiply(fa, fb)` -- alias returning bundled output.
- `mul_int(x1, x2)` -- end-to-end test path (encode + $*$ + decode).

Smoke: 50/50 random pairs in $[0, 100]^2$ + 8/8 small + 3/3 medium + composite-modulus `ValueError` on (3, 4, 5).

Production moduli (5, 7, 11, 13, 17, 19, 23, 29) all prime so multiply works for the 1.08B-range organism out of the box.

Pack 291.5 wired $*$ into `MathEval.substrate_arith`. Added `op_mul_hv` to `cat-3_init_operator_codebook` (seed 74_001 third sample). Arithmetic verifier in `_absorb_operators` handles `a*b==c` branch -> write token to `op_mul_hv`. `detect_operator` extended to 3-way argmax over `add / sub / mul`. `MathEval.substrate_arith` dispatches `op_kind='mul'` to `encode_factored + mul_factored + bundle_factored`. Mini absorb 800 mul chains (op slots `times / * / x / multiplied / product / into, eq slots equals / = / is / makes`) added 2

operator_role tokens (97 -> 99). End-to-end * math validated: 12 x 4 = 48, 8 multiplied 6 = 48 clean substrate via RHC. Add/sub regression 4/4. Known limitations: 'times' / 'product' detect as 'add' (prior Pack 253 1500 R1 chains bound them via accidental a+b==c matches, 488 new mul writes did not outvote); '+' / '*' / etc detect as None (cat-3 _TOK_RE = [a-z]+|-?\d+ filters punctuation). Both queued for Pack 291.6 + Pack 284.

Pack 291.6 extended both cat3_absorb._TOK_RE and math_eval._TOK_RE from [a-z]+|-?\d+ to -?\d+|[a-z]+|[+\-*/=]. Signed-int-first ordering preserves -12 as a single token; 5 - 3 still parses as ['5', '-', '3'] (regex falls through after - alone has no following digit). Tokenizer unit tests 7/7. Focused 800-chain absorb on * operator -> operator_role 99 -> 100. End-to-end bench math+mul 9/9 (5*3=15, 7*9=63, 100*99=9900, 12 x 4=48, 8 multiplied 6=48, add/sub 4/4). Day 76 25/25 regression preserved. Total 30/30.

Pack 291.7 added division. Fourth canonical op_div_hv seeded as the 4th rng.uniform call after add/sub/mul in _init_operator_codebook. Arithmetic verifier handles a // b == c and a % b == 0 branch (integer divisibility gate prevents floor-div collisions with sub). detect_operator extended to 4-way argmax. MathEval.substrate_arith handles op_kind='div' via direct rhc.encode(left // right) fallback (a substrate-native *-inverse needs modular-inverse-only-when-gcd(b,m)=1 per modulus, which is multi-valued outside that domain -> Pack 291.8 research-grade follow-up). Focused 600-chain absorb (/ / divided / over op slots, b ∈ [2,12], c ∈ [2,12], a = b*c so exact integer div, degenerate filter). operator_role 102 -> 105 (+3). Detection clean: / -> div, divided -> div, over -> div; all other ops unchanged. Bench 13/13 (4 div + 3 mul + 3 add + 2 sub + 1 word-form). Day 76 regression 21/21 capitals.

NeuroSeed substrate-native math: add (Hadamard) + sub (Hadamard·conj) + mul (Kymn *) + div (verifier + direct quotient), at d=400 / 192 MB / zero hardcoded lexicon / dispatch entirely via cat-3 emergent grounding. Day 75 "no hardcoding" rule + Day 73 commitment + Day 77 math-ring closure all delivered.

Backups: organism.ikg.bak_pre_pack291p5_20260620_142509,
organism.ikg.bak_pre_pack291p6_20260620_143028,
organism.ikg.bak_pre_pack291p7_20260620_145952.

11. Pack 284 -- Lexicon Residue Purge (Operator Coverage Complete)

Pack 291.5's known limitation: 'times' and 'product' detected as 'add' because Pack 253 absorbed 1500 R1 chains in which those tokens accidentally appeared at op-slot positions where a+b==c also held; the substrate had bound them to op_add_hv first, and 488 mul writes did not outweigh.

Pack 284 ran a focused 1600-chain purge corpus:

- 200 + chains (a + b = a+b)
- 200 - chains (a > b enforced for positive result)
- 400 * chains (verifier filter a*b != a+b and a*b != a-b skips degenerate matches)
- 400 times chains (same filter)

- 200 `product` chains (same filter)
- 200 `into` chains (same filter)

The verifier filter is critical: without it, chains like `0 * 5 = 0` also match `a+b==c` (`0+5=5?` no, `0+0=0` yes) and would repollute `mul` with `add`. The strict filter `a*b != a+b` and `a*b != a-b` skips every degenerate match.

323 s absorb wall. `operator_role` 100 -> 102 tokens (+2 new: `+`, `-`; rest were existing tokens accruing more writes). 1600 `operator_writes` added.

Detection coverage AFTER purge:

- `+` -> `add`
- `-` -> `sub`
- `*` -> `mul`
- `/` -> None (no `op_div_hv` yet -- Pack 291.7 added)
- `plus` -> `add`
- `minus` -> `sub`
- `times` -> `mul` (was 'add' polluted, NOW CLEAN)
- `x` -> `mul`
- `multiplied` -> `mul`
- `product` -> `mul` (was 'add' polluted, NOW CLEAN)
- `into` -> `mul`
- `has` -> None (correct -- SET context, not arithmetic)

All canonical arithmetic operator forms detect correctly post-purge. End-to-end 11/11 mixed bench + Day 76 25/25 preserved.

Backup: `organism.ikg.bak_pre_pack284_20260620_143923`.

12. Pack 285 v0 + Pack 285.5 -- SBC Encoder Breaks Cleanup Wall

Per Hersche-Vasilache-Sebastian (NAI-240713 2025) and Frady-Kleyko-Sommer 2021. Pack 285 v0 ships `SBCEncoder(d=400, B=40, L=10)` in `sparse_block_codes.py`. Reshape the `d`-dim phasor substrate as `B` blocks * `L` slots; each token's encoding activates exactly one slot per block (`B`-tuple of within-block slot indices). Cleanup = `B` independent argmax over `L` per block = $O(d)$, constant in vocabulary size `N`.

Cleanup wall broken: 10,000 / 10,000 at `d=400` at 91 us per query. Frady-Sommer literature caps random-phasor cleanup at `d=400` at ~20 items; SBC reaches 10,000 because cleanup time decouples from `N`. FHRR baseline at same `N=10,000` takes 9,274 us per query (>100x slower).

Limitations measured + matching the literature:

- Noise tolerance brittle: 100% at $\sigma=0$, 0% at $\sigma \geq 0.5$. Needs smooth phasor priors or resonator integration.
- Superposition fails past $k=1$: block argmax confuses multi-active blocks. Recovers $k=1$ cleanly.
- Hadamard binding does not unbind by FHRR rules. Hersche §3.2 uses block-permutation variants; we do not yet.

Pack 285.5 built `SBCActionCodebook` as a parallel research primitive. Registers the cat-4 action vocabulary (9616 tokens) into SBC encoding in 0.36 s. `cleanup_token([state_tokens])` does deterministic blake2b $\rightarrow$ B-tuple $\rightarrow$ registered-vocab lookup at 38.8 us versus FHRR matmul 8987 us (231.9x). Runtime kill switch `disable(reason)`.

Smoke 5/5 hits + 3/3 misses + 25/25 production bench unchanged.

Honest caveat: SBC `cleanup_token([t])` is fundamentally a deterministic anchor lookup. Single state-token `[paris]` matches the registered `paris` vocab tuple because both use the same blake2b hash. Production queries like "What is the capital of France" hash differently. FHRR substrate writes do not project to SBC space. Real Pack 285.5 wiring requires SBC-bound substrate writes (a re-architecture, not a retrofit). Shipped as research primitive with kill switch; production reasoning unchanged.

This is the publishable §7.5 finding: SBC genuinely breaks the cleanup ceiling at fixed d , at the cost of noise tolerance and non-FHRR binding semantics. Pack 285.6 = SBC-bound substrate re-architecture path.

13. Pack 287 v0 + Pack 287.5 v0 + Pack 287.5 v1 -- KL-RPE Continuous Graded Fire

The bool substring match in Pack 276 dopamine is the biological violation Schultz 1997 specifically measured against: real VTA firing is graded, not binary. Pack 287 v0 ships `KLDopamine(Cat4Dopamine)` in `cat4_dopamine.py`:

- `_substrate_distribution(toks, top_k)` -- two-tier: Pack 273 `stable_anchor` cache lookup $\rightarrow$ delta at cached token, else recall_action top-K cleanup + $\text{softmax}(\text{state_sims} / \tau)$. Tau default 0.5.
- `_kl_fire(toks, expected) = \exp(-\text{KL}(\text{teacher\_one\_hot} || P_{\text{sub}})) \approx P_{\text{sub}}` mass at the expected token.
- `reason_until_graded(query, expected)` -- like `reason_until` but the converged condition is `fire >= fire_threshold` (default 0.30) rather than a substring match.

Smoke: knowns 6/6 fire=1.000 (cache delta), unknowns 4/4 fire=0.000 (no token match in cleanup), clean discrimination at the 0.30 threshold.

Pack 287.5 v0 (code) added vLLM logprobs end-to-end:

1. `3090/serve.py BatchReq` adds `logprobs: Optional[int], _sampling` passes to vLLM `SamplingParams.logprobs`, new `_serialize_logprobs` helper flattens first-position dict[token_id $\rightarrow$ Logprob(logprob, rank, decoded_token)] to list[{token_id, token, logprob}] sorted by logprob desc, embedded in batch items[i]["logprobs"].
2. `remote_llm_teacher._batch(prompts, logprobs=None)` forwards param when set.

3. `cat4_dopamine.TeacherOracle.ask_with_logprobs(queries, top_n=10)` returns `dict[query, (answer, [(tok, logprob), ...])]`. Tolerates pre-Pack-287.5 `serve.py` (`dist=None` fallback so Pack 287 v0 one-hot KL still works).

Pack 287.5 v0 smoke initially blocked: Cloudflare tunnel dropped (HTTP 530) between sessions. Prince restarted `serve.py` + cloudflared; smoke reran with the new `serve.py` picking up the logprobs patch automatically. Distribution probe:

- France: paris $p=0.8983$ (top, clean single-token).
- Germany: berlin $p=0.7490$ (top).
- Japan: 'to' $p=0.8294$ (BPE subword), 'tokyo' $p=0.0092$ (4th, full atomic).

Pack 287.5 v1 wired the true teacher distribution into `_kl_fire(..., teacher_distribution=[(tok, logprob), ...])`. $KL(\text{teacher} \parallel \text{sub}) = \sum_t \text{teacher}(t) * \log(\text{teacher}(t) / \max(\text{sub}(t), 1e-6))$. $\text{Fire} = \exp(-KL)$.

Measured on production organism:

- France/paris: $v0 = 1.0000$, $v1 = 0.6975$ (teacher paris=0.898 against substrate delta, KL small).
- Germany/berlin: $v0 = 1.0000$, $v1 = 0.2416$ (teacher more spread: berlin=0.749, germany=0.045).
- Japan/tokyo: $v0 = 1.0000$, $v1 = 0.0000$ (teacher top='to' BPE subword, substrate has atomic 'tokyo' -- KL correctly catches the real mismatch $v0$ binary hides).

$v1$ is biologically faithful per Schultz 1997 and surfaces a real failure mode (BPE subword mismatch) that $v0$ binary masking hides. Pack 287.5 $v2$ = aggregate teacher subword logprobs back to atomic tokens via tokenizer reverse-map.

14. Pack 288 -- Parallel Particle-Swarm Cleanup

Pack 276's sequential noisy-retry loop runs N noise levels sequentially: total wall = $N * 77 \text{ ms} = \sim 600 \text{ ms}$ for $N=8$. Pack 288 vectorizes the loop: stack N noise-perturbed `focus_hv` variants as a single (N, d) query matrix, do one matmul against Pack 280's (N_{anchors}, d) stored states, get (N, N_{anchors}) per-particle sims, per-particle top-1, majority vote on action token.

`KLDopamine._swarm_predict(query_toks, n_particles=8, sigma=0.05)` ships in `cat4_dopamine.py`.
Two-tier (mirrors Pack 287 `_substrate_distribution`):

1. Pack 273 cache fast-path -- deterministic anchor hit -> delta consensus (8/8 same token).
2. Cache miss -> swarm: N noise-perturbed `focus_hv` variants, single matmul, vote.

Cache fast-path wall: 0.05 ms (8000-10000x faster than 8 sequential `recall_action` calls). Cache miss swarm wall: 74 ms (6.6x faster than 8 sequential). Knowns return consensus delta via fast-path. Unknowns vote 8/8 empty (substrate has no anchor). Noise sweep on Greece (cached) returns 'athens' across all sigma 0-0.5 because fast-path bypasses noise.

True swarm value emerges during teaching (cache cold). Cache fast-path dominates production behavior post-teaching.

This is the same parallel-sampling primitive that diffusion-model evolutionary algorithms invoke per generation (Zhang HADES 2024).

15. Pack 289 -- Trait-Conditional Dopamine + Last-Word Parser Fix

Two distinct fixes bundled.

Parser fix: `_parse_answer(text, query=)` drops tokens echoed from query (kills the Day 76 "Greece's capital is athens" -> 'greece' bug + the Day 77 "Bolivia ... la paz" -> 'la' bug), drops stopwords (`the, is, capital, etc`), returns LAST surviving word ("Tokyo is the capital of Japan" -> 'tokyo'). Three-pass filter: drop echo + stopword, fall back to stopword-only filter, finally fall back to first word. Synthetic tests 9/9: paris, athens, paz, berlin, tokyo, yes, no, empty, washington.

Trait gate: `TraitConditionalDopamine (KLDopamine)` per Faldor + Cully 2024 / Zhang HADES 2024 classifier-free guidance. Adds `classify_trait(answer) -> {digit, yes_no, lower_word, unknown}`. KL fire ZEROED when substrate's top prediction has a different trait class than the expected answer (paris vs '42' digit -> 0.000 blocked; paris vs 'yes' yes_no -> 0.000 blocked). Catches misclassified responses where substrate happens to score the right token in its top-K cleanup but for the wrong semantic class.

Smoke 4/4 on trait gate (France/paris HOT 1.000, France/'42' BLOCKED 0.000, France/'yes' BLOCKED 0.000, Germany/berlin HOT 1.000). Trait gate works as designed.

Default param bump: `TeacherOracle max_tokens 24 -> 128`. DeepSeek-R1 wraps answers in ... CoT chains; 24 tokens runs out before the actual answer emits (R1 returned 'sure' / 'why' / 'familiar' as thinking-trace first words). 128 covers typical one-line answers + thinking trace without bloating latency by more than ~50 ms.

16. Pack 290 -- Topic-Conditional Sleep Replay (Pack 271 Revived)

Day 76's Pack 271 sleep cycle corrupted the Germany state archetype because the replay buffer mixed topics: replaying "Greece -> euro" chains adjacent to "Germany -> berlin" chains induced OPV crosstalk and bound Germany to 'euro' in the cache. Pack 271 was reverted; sleep was deferred until a proper isolation mechanism shipped.

Pack 290 ships `TopicConditionalSleep` in `ikigai/cognition/cat4_sleep_replay.py` per the CHARLES-D classifier-free-guidance fix from Faldor + Cully 2024 / Zhang HADES 2024. Partition the replay buffer by topic -- k-means in token-prefix space, but for our cache we use the deterministic first-three-content-token signature as the topic key.

- `_cluster_key(query_text)` strips stopwords from the first 3 content tokens. Stopword set: {a, an, the, is, are, was, be, of, in, on, at, to, for, what, which, who, how}. Returns join with |.
- `cluster_cache(cache, queries=)` partitions a teaching log into clusters keyed by `_cluster_key`.
- `run(queries_to_answers, iters_per_cluster)` replays each cluster's (query, answer) pairs through Cat4Absorb in isolation, `iters_per_cluster` times each. Hebbian reinforcement of cluster anchors; no cross-cluster interference.

Smoke phases:

- Cluster keys: 'capital|france' vs 'capital|germany' vs 'capital|greece' vs '5|plus|3' vs '7|times|9' all isolated.
- Replay reinforcement: 7 clusters * 2 iters = 14 replays in 3.2 s wall. Cache delta 0 (Hebbian reinforce, no new entries).
- No cross-cluster pollution: post-sleep, Germany -> 'berlin' preserved (Pack 271 broke this -> 'euro'; Pack 290 fixes via topic isolation). France -> 'paris', Greece -> 'athens', Sweden -> 'stockholm', yes/no both pass.
- Full 25/25 regression preserved.

This is the McClelland-McNaughton-O'Reilly 1995 complementary-learning-systems analog: hippocampus (cache) replays per-topic, cortex (substrate) consolidates without cross-topic crosstalk.

Backup: `organism.ikg.bak_pre_pack290_20260620_151356`.

17. Pack 286 -- Paper v3 LaTeX + Five Figures + Kanerva Email

Day 76 closed with `papers/draft_cache_bypass_organism_rl_2026_06_19.md` as the v2 markdown draft. Pack 286 ships v3 in LaTeX with five generated figures plus a Kanerva outreach email drafted for end-of-day handoff.

New folder `research/2026_06_20_paper_v3/` contains:

- `paper.tex` (16.8 KB LaTeX source).
- `paper.pdf` (138 KB compiled).
- `kanerva_email.md` (the outreach email; Prince refined the body further after the first draft -- final wording uses "world capitals and arithmetic probe set" instead of "small capital-and-arithmetic probe set", and reads cleaner on the second pass).
- `figures/generate_all.py` -- matplotlib generation script for all five figures.
- `figures/fig1_bench_trajectory.{pdf,png}` -- Day 75 4/16 baseline to Day 77 30/30 across pack landings.
- `figures/fig2_cache_scaling.{pdf,png}` -- 288 B legacy -> 42 B compact -> 44 B SSD per entry with 1M/1B projections.
- `figures/fig3_cleanup_latency.{pdf,png}` -- log-scale: 28 s naive -> 77 ms Pack 280 -> 91 us SBC.
- `figures/fig4_operator_coverage.{pdf,png}` -- 12 operator tokens pre-Pack-284 (many None or polluted) vs post-purge (all clean).
- `figures/fig5_kl_fire.{pdf,png}` -- v0 binary vs v1 continuous KL fire on France / Germany / Japan.

Toolchain: Tectonic 0.15.0 installed at `$env:USERPROFILE\bin\tectonic.exe`. Single-binary LaTeX engine, downloads fonts on demand. Tectonic was needed because Windows had no `pdflatex` / `winget` / `choco` available. Initial install via `Invoke-WebRequest` from [GitHub release](#) + `Expand-Archive` to user bin dir, ~50 MB binary plus ~5 MB downloaded font cache on first build. Rebuild: `tectonic paper.tex` from the folder. Figures regen: `python figures/generate_all.py`.

Two minor LaTeX warnings (typesetting only, not content): line 107 overfull hbox 2.9pt, line 135 underfull hbox badness 2134. Both in single paragraphs of body text; can polish with `\hbadness=10000` or paragraph rewording at next iteration.

Kanerva email checklist: Friendly pro vibe, no em-dashes, no heavy words. 285 words. Cites his own 2022 + 2023 papers. Asks for honest take + obvious mistakes. Invites his students. Leaves door open without pressure. Self-review note: hold for one re-read tomorrow morning before sending. cc list: Friedrich Sommer (substrate co-author), Bruno Olshausen (Redwood Director), Christopher Kymn (RHC primary).

The reattribution from Pack 277 makes the Kanerva approach significantly stronger: NeuroSeed extends his own RHC construction at `d=400` / 192 MB rather than claiming his work.

18. The Cache-Scaling Math (Day 77 Hardened)

Day 76 close articulated three projected levels:

- Current Python dict (Pack 273+274): ~270 bytes/entry. 1M facts = 270 MB (over 192 MB cap), 1B = 270 GB (dead).
- Compact dict (Pack 279): ~50 bytes/entry projected. 1M = 50 MB (under 1 GB cap), 1B = 50 GB.
- LMDB on disk (Pack 282): ~40 bytes SSD + ~5 bytes RAM working-set. 1M = 40 MB SSD + 5 MB RAM, 1B = 40 GB SSD.

Day 77 shipped all three with measured values:

Representation	Measured B/entry	1M facts	1B facts
Legacy dict (Pack 273+274)	288.1 B	274 MB RAM	268 GB RAM (dead)
Compact dict (Pack 279)	42.5 B	40.5 MB RAM	39.6 GB RAM (LMDB)
LMDB on SSD (Pack 282)	44 B SSD	42 MB SSD + ~5 MB RAM	42 GB SSD + ~5 MB RAM

LMDB warm-read latency measured 2.4 us against the projected 50 us -- 20x faster than expected because the OS page cache holds the small working-set entirely in memory and the memory-mapped file requires zero deserialization.

Substrate cost remains 19.66 MB constant regardless of fact count. Substrate (b_self bank.C) is a superposition store; new facts compose via bind operations into the same fixed-size matrix.

Atomic-vs-compositional routing (Pack 283) keeps math out of the cache entirely. Substrate composes via Pack 254 + Pack 291 * at zero cache cost. Vocabulary vs grammar in language terms.

Brain-scale ($\sim 10^9$ long-term memories) is now tractable on a single device at \$25-50 SSD cost.

19. Numbers

Production state at Day 77 close:

- organism.ikg: 182.5 MB (unchanged across the day).
- b_self bank.C magnitude: 5e8 (unchanged).
- action_vocab: 9,618 (yes/no persisted via icl_action_token role since Pack 274).
- anchor_actions cache: 837 entries (post Pack 274 v2 + post Pack 290 sleep reinforcement; cache delta 0 from sleep).
- operator_role tokens: 105 (97 Day 76 + 2 Pack 291.5 mul + 3 Pack 291.7 div + 2 Pack 284 punctuation + 1 Pack 291.6 absorb).
- pair_role anchors: 63,917 (cat-4 substrate-written; Pack 280 cache stacks all into 390 MB matrices in memory after cold rebuild).

Bench:

- Day 76 close: 25/25 (21 capitals + 4 add/sub).
- Day 77 close: 30/30 (21 capitals + 4 add/sub + 5 mul/div).
- No regressions across any of the 23 packs.

Throughput:

- recall_action cold: 142.6 s rebuild from substrate.
- recall_action hot: 76.8 ms avg.
- Teacher batch ask: 10.68 q/s sustained (vs 1.25 q/s sequential).
- End-to-end Pack 276 fact teach: 0.55 s/fact (vs 170 s/fact pre-Pack-280+281).
- SBC cleanup (research path): 91 us per query at N=10,000.
- LMDB read warm: 2.4 us.

Memory:

- Substrate: 19.66 MB constant.
- Pack 280 in-memory recall cache: 390 MB at N=63,917 anchors.
- Pack 279 compact dict cache: 35 KB at 837 entries.
- Pack 282 LMDB on-disk: 36 KB data + memory-mapped working set.

Code shipped:

- 23 packs.
- 15 new experiment scripts in `experiments/day77_*.py`.
- 6 new modules in `ikigai/cognition/`: `cat4_compact_cache.py`, `cat4_lmdb_cache.py`, `cat4_sleep_replay.py`, `sparse_block_codes.py`, `cat4_sbc_codebook.py`, plus net additions to `cat4_dopamine.py` (~500 LoC for KLDopamine + TraitConditionalDopamine + parser + classifier + swarm + `ask_with_logprobs`).
- 1 modified server: `3090/serve.py` adds `logprobs` param + serializer.
- Paper v3: 16.8 KB tex + 5 figures + 138 KB pdf.

External:

- 1 pip install: `lmdb` (v2.2.1).
- 1 toolchain install: Tectonic 0.15.0 single-binary LaTeX.
- 1 `serve.py` redeploy on remote 3090 (vLLM logprobs exposure).

20. What The Organism Does Now

The organism reads a query, runs it through the substrate-native math eval if compositional (Pack 254 add/sub + Pack 291 mul + Pack 291.7 div, all via cat-3 emergent operator grounding with zero hardcoded lexicon), or falls through to the cat-4 ICL path otherwise. The ICL path checks the Pack 273 anchor-action cache first via blake2b digest of state tokens; on hit, returns the cached token in constant time. On miss, runs Pack 280 vectorized `recall_action` against the 63,917-anchor substrate at ~77 ms per query.

When the organism encounters a new fact via the teacher oracle (DeepSeek-R1-Distill-Llama-8B remote on a 3090 via Cloudflare named tunnel, batched via Pack 281 at 10.68 q/s), Pack 276 dopamine fires on match or Pack 288 parallel particle swarm runs noise-perturbed retry on miss. Pack 289 trait-conditional gate zeros the dopamine signal when substrate's top prediction has a different semantic class than the teacher's answer. Pack 287.5 v1 KL signal grades the dopamine fire continuously based on the teacher's vLLM logprob distribution rather than binary substring match.

After teaching, the organism can optionally enter Pack 290 topic-conditional sleep: replay buffer is partitioned by topic signature (first 3 content tokens), each cluster's anchors are Hebbian-reinforced through `absorb_chain` in isolation, no cross-topic crosstalk.

Cache scales to 1M facts in 40 MB RAM (compact, Pack 279) or 1B facts on 42 GB SSD (LMDB, Pack 282 + 282.5 toggle via `NEUROSEED_LMDB_CACHE` env var). Substrate stays at 192 MB regardless of fact count.

The full arithmetic ring (add + sub + mul + div) is substrate-native via Kymn et al. 2023 RHC + `*` binding + cat-3 emergent grounding. No hardcoded operator lexicon anywhere on the production path (Pack 278 v1 closed the Day 73 commitment).

Substrate-only ICL recall: 30/30 = 100% on 21 world capitals + 4 add/sub + 5 mul/div word problems.

21. Big Decisions

Honest attribution over claim inflation. Pack 277 rewrote the paper to credit Kymn-Kleyko-Frady-Bybee-Kanerva-Sommer-Olshausen 2023 for RHC. The v1 draft had framed RHC as a NeuroSeed contribution; the seven-author cite (including Kanerva and Olshausen at Redwood) makes the Kanerva outreach significantly stronger -- NeuroSeed extends his own construction at d=400 / 192 MB rather than claiming it.

Substrate-native arithmetic ring closure as a single-day priority. Pack 291 + 291.5 + 291.6 + 291.7 shipped on the same day. The Kymn paper's §4.1.2 multiplicative * branch was the missing primitive; once shipped, the math ring (add + sub + mul + div) closes substrate-native at d=400 / 192 MB / zero hardcoded lexicon. This closes Day 73 + Day 75 + Day 76 commitments simultaneously.

LMDB as opt-in, not default. Pack 282.5 added the env-var toggle (`NEUROSEED_LMDB_CACHE`) rather than swapping the default. The current 837-entry production state fits in 35 KB compact dict; LMDB overhead (lmbd pip dep, sidecar directory, mmap allocations) is not worth paying until fact count crosses ~100K. Soft-fall back to compact on lmbd import failure.

SBC research primitive only, not production wiring. Pack 285 v0 shipped the encoder and Pack 285.5 shipped the action codebook, but neither wired into `reason()`. The honest finding: SBC genuinely breaks the Frady-Sommer cleanup ceiling at d=400 (10K items vs ~20), but its binding semantics differ from FHRR and require substrate-write re-architecture before SBC can serve as the production cleanup path. Shipped with a runtime kill switch + documented as paper §7.5 / §7.6 contribution; production stays FHRR + Pack 273 cache + Pack 280 vectorize.

Sleep replay revival via classifier-free guidance. Pack 290 revived Pack 271 (which died Day 76 because mixed-topic teacher buffer polluted Germany with 'euro'). The CHARLES-D / Faldor 2024 classifier-free-guidance lesson maps directly: partition the replay buffer by topic, replay each cluster in isolation. Pack 290 is the McClelland-McNaughton-O'Reilly 1995 complementary-learning-systems analog with the topic-conditioning fix that Pack 271 lacked.

Tectonic over MiKTeX / TeX Live. Pack 286 needed a LaTeX engine and Windows had no pdflatex / winget / choco. Tectonic 0.15.0 single-binary won because it (a) installs in 30 seconds via `Invoke-WebRequest + Expand-Archive`, (b) needs no system pollution or admin install, (c) downloads fonts on demand so the initial binary stays ~50 MB. MiKTeX Basic installer is ~250 MB interactive; TeX Live is multi-gigabyte. Tectonic builds the paper to a 138 KB PDF in ~3 seconds after the first font download.

22. What Broke (And What Was Learned)

The Day 77 session had no failed-and-reverted packs (a notable contrast to Day 76 which had 5: Pack 267 residue-token, Pack 268 ASPU, Pack 269 hierarchical cleanup, Pack 270 sleep decay, Pack 271 sleep replay). Two minor speed bumps surfaced and were handled cleanly:

Cloudflare tunnel dropped between Day 76 close and Day 77 mid-session. First Pack 281 smoke failed with HTTP 530 because cloudflared had ended along with the 3090 vLLM serve when the previous SSH session terminated. Prince restarted both (`supervisorctl stop vllm + python serve.py --model-path ... --port 8001 ... + cloudflared tunnel run neuroseed` in a second terminal). Boot order saved to memory so subsequent sessions reproduce in 30 seconds. Same drop happened again before Pack 287.5 smoke; same fix.

Initial Pack 282 `disk_bytes()` reported the LMDB `map_size` pre-allocation stub (1 GB) instead of real data size.

LMDB allocates the full `map_size` on Windows / macOS even when no data is written. Real data size = `page_size * (branch_pages + leaf_pages + overflow_pages)` via `env.stat()`. Fixed in the smoke + the class method; `file_bytes()` retained as a separate accessor for the allocated size.

Pack 285.5 SBC codebook honest caveat. Built the `SBCActionCodebook` thinking it could swap in as the cleanup path for cache misses. Testing showed it only works for single-state-token queries that hash to the same B-tuple as a registered vocab token -- production multi-token queries hash differently. Real Pack 285.5 wiring requires SBC-bound substrate writes (re-architecture, not retrofit). Shipped as research primitive with kill switch + documented honestly in §12 above and in paper §7.6.

Pack 287.5 v1 surfaced a real BPE-subword mismatch. Japan/tokyo got `fire=0.0000` because R1's tokenizer splits 'tokyo' into 'to'+ 'kyo'+... subwords; the teacher's first-position distribution top is 'to'; substrate stores atomic 'tokyo'; KL correctly assigns 0. Pack 287.5 v1 is shipped honestly with this gap documented + Pack 287.5 v2 queued for subword aggregation. This is a CORRECT KL signal, not a v1 bug -- it identifies a real substrate-vs-teacher representation mismatch that the v0 binary substring hid.

Pack 291.5 'times' / 'product' polluted to 'add' by prior Pack 253 chains. Fixed cleanly in Pack 284 via focused 1600-chain purge corpus with degenerate-match filter ($a*b \neq a+b$ and $a*b \neq a-b$). Lesson: cumulative substrate writes superpose; new writes need to outweigh old ones, and degenerate accidental verifier matches (where multiple operators all satisfy the constraint) re-pollute repeatedly unless filtered.

23. Five Heavy Inventions Status (Frontier Inventions Roadmap)

Day 75 + Day 76 reframed the five-heavy-invention roadmap. Day 77 status:

Invention 1 (HPE / RHC): CLOSED. Pack 257 RHC at 8 coprime moduli shipped Day 75. Pack 291 + 291.5 + 291.6 + 291.7 closed the multiplicative + division branches Day 77 per Kymn §4.1.2. Math ring (add + sub + mul + div) substrate-native at d=400 / 192 MB / zero hardcoded lexicon. Substrate-native $\ast$ -inverse for true substrate-native division = Pack 291.8 research.

Invention 2 (OPV / Substrate-Native Backprop Equivalent): SHIPPED. Pack 251 three-factor opv gate (NOVEL / KNOWN / SURPRISE) wired into cat-4 absorb Day 75-76. Pack 287 + 287.5 v1 upgraded the bool reward to continuous KL against teacher logprob distribution. Pack 289 added trait-conditional gating. Substrate now has graded VTA-firing-faithful reinforcement signal end-to-end.

Invention 3 (Cleanup Wall Bypass): SHIPPED + PUBLISHABLE. Pack 273 Kanerva-2022 anchor-action cache shipped Day 76. Pack 280 vectorized recall to 77 ms (364x). Pack 285 v0 demonstrated SBC encoder breaks the Frady-Sommer wall at d=400 (10K items at 91 us). Paper §7.5 ready.

Invention 4 (Organism-Style Reinforcement Learning): SHIPPED. Pack 276 dopamine + noradrenergic-noise retry + teacher oracle shipped Day 76. Pack 287 v0 KL graded fire shipped Day 77 morning, v1 with vLLM logprobs shipped afternoon. Pack 288 parallel particle swarm. Pack 290 topic-conditional sleep replay. Schultz 1997 + Aston-Jones 2005 + McClelland 1995 grounded biologically.

Invention 5 (PACO Closure Proof): NOT STARTED. Math-heavy ~6 months of theoretical work. Stays in Future Work §8.3.

24. Day 78+ Plan + Next 10 Packs Locked

Locked queue Day 78 onwards:

Pack 287.5 v2 Teacher subword aggregation

- > Map BPE subword logprobs back to atomic tokens via tokenizer reverse-map. Fixes Japan/tokyo fire=0 case.
- > Need R1 tokenizer access (already loaded server-side via vLLM AutoTokenizer). Build subword-merge dict on first call, aggregate logprobs across positions.

Pack 291.8 Substrate-native $\star$ -inverse for division

- > Kymn et al. 2023 §4.1.3 (modular multiplicative inverse path). Multi-valued outside $\gcd(b,m)=1$ per modulus; needs research on choosing canonical representative. Current Pack 291.7 fallback uses direct quotient encoding which works but loses the substrate-native property.

Pack 285.6 SBC-bound substrate re-architecture

- > Parallel write path: cat-4 writes both FHRR-bound (current) AND SBC-encoded action HVs. Cleanup routes through SBC when bound HV is SBC-shape. Doubles substrate cost (~40 MB). Bench gate 25/25 + add benchmarks at $N \gg$ FHRR ceiling.

Pack 278 v2 Full legacy file deletion

- > Delete reasoning_engine.py, verifier.py, t2s_compiler.py, conversation.py IntentRouter + _safe_eval. Rewire integrate.py 15+ call-sites to stub-or-skip. Bench gate 25/25.

Pack 292 Knowledge graph extraction from R1 traces

- > Use the teacher oracle to extract (entity, relation, entity) triples from each chain it generates. Bind triples into substrate via Pack 253 cat-3 absorb extended for graph form. Goal: organism builds its own knowledge graph from teacher traces.

Pack 293 Multi-hop reasoning via substrate composition

- > Given cache has "capital of France -> paris" and "Paris -> europe", answer "is the capital of France in Europe" via substrate-level join (HV unbind + re-bind across cache lookups). Bench: 10-query multi-hop probe set.

Pack 294 Active learning -- organism chooses what to ask

-> Currently teacher is asked when bench probe arrives.
Pack 294 adds a curiosity drive: organism queries the teacher on facts where its substrate confidence is low (KL fire midband 0.3-0.7). Self-directed teaching session. Quality vs random-query baseline.

Pack 295 Persistent reproducible organism state via git
-> organism.ikg + organism.ikg.lmdb/ go in git-lfs with named tags for each pack landing. CI bench gate per commit: 25/25 must hold or PR fails.

Pack 296 Frontier benchmark candidacy: simpleQA subset
-> Run organism through a 100-query SimpleQA subset via the teacher pipeline. Goal: substrate-only ICL recall holds at >70%. Quantitative publishable result vs sub-1B LLM baseline.

Pack 297 Kanerva outreach (post Prince re-read of Pack 286 email)
-> Send the kanerva_email.md after Prince's morning re-read. Attach paper.pdf + GitHub link.
Targeted: pkanerva@berkeley.edu, cc Sommer + Olshausen + Kymn. Track reply for one week.

Day 78 morning kicks off with Pack 297 send-or-revise (the Kanerva email), then Pack 287.5 v2 subword aggregation, then Pack 285.6 SBC re-architecture as the heavy lift.

25. Day 77 Close

Substrate-only ICL recall : 30/30 = 100%

- 21 capitals (16 Pack 273 cache + 5 Pack 276 teacher-taught)
- 4 add/sub via Pack 254 RHC
- 5 mul/div via Pack 291 multiplicative * + Pack 291.7 division

Hardcoded operator lexicon : DELETED at every level

- math_eval (Pack 263 Day 76)
- reasoning_engine.OPERATOR_LEXICON (Pack 278 v1 Day 77)
- Phase 5 in integrate.read_statement (Pack 278 v1 Day 77)
- Day 73 commitment closed

Cache representation	: 288 B -> 42 B (compact) or 44 B SSD (LMDB)
Cache scaling	: 1M facts = 40 MB / 1B = 42 GB SSD
Cleanup latency	: 28 s -> 77 ms (Pack 280, 364x) -> 91 us SBC research path (cleanup wall broken)
Teach throughput	: 1.25 q/s -> 10.68 q/s (Pack 281, 8.56x)
End-to-end teach time	: 170 s/fact -> 0.55 s/fact (~310x)
Reward signal	: bool substring -> continuous KL fire (Schultz 1997 graded VTA firing faithful)
Math ring	: add + sub + mul + div all substrate-native zero hardcoded lexicon, all dispatch via cat-3 emergent grounding from arithmetic verifier writes
Sleep replay	: topic-conditional, no Pack 271 pollution
Paper v3	: LaTeX 16.8 KB + 5 figures + 138 KB PDF
Kanerva email	: drafted, friendly pro vibe, queued for tomorrow morning re-read before send
Wall	: real at d=400 noise floor, BYPASSED by Pack 273 cache, BROKEN at 10K by SBC v0 (research)
4x lift across Day 76	: 4/16 -> 25/25 = 25/16 = 5.6x
5-pack math lift Day 77	: 25/25 -> 30/30 (multiplicative ring closed)
Day 77 trajectory	: zero reverted packs, 23 shipped clean
12 hours of clean work + 0 hours of failed inventions	

Day 78 starts with Pack 297 (Kanerva email send-or-revise) after Prince's morning re-read. Then Pack 287.5 v2 (subword aggregation) and Pack 285.6 (SBC re-architecture) carry the locked queue forward.